

EXERCISE-4

Design Intrusion Detection System

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CLASS: IOT-B

AIM:

Create a security system that alerts when motion is detected.

COMPONENTS REQUIRED:

SNO	COMPONENT	NO'S
1	LED	1
2	PIR MOTION SENSOR	1
3	BUZZER	1
4	ARDUINO UNO BOARD	1
5	RESISTORS	2
6	BREAD BOARD	1

FEATURES:

Detects motion and activate a buzzer or light as an alarm.

METHODOLOGY:

- Use a Pir motion sensor available in the tinker cad library
- Connect the Pir sensor to a digital pin on Arduino and a buzzer / LED for alerts
- Write a simple Arduino program to activate the buzzer or LED when detecting motion.

CONNECTIONS:

PIR Motion Sensor Connections

- VCC (Power) → Arduino 5V
- GND (Ground) → Arduino GND
- OUT (Signal) → Arduino Digital Pin 2
-

LED with Resistor (220Ω – 1KΩ)

- Arduino Digital Pin 4 → One leg of 220Ω – 1KΩ resistor
- Other leg of resistor → LED Anode (+, long leg)
- LED Cathode (-, short leg) → Arduino GND

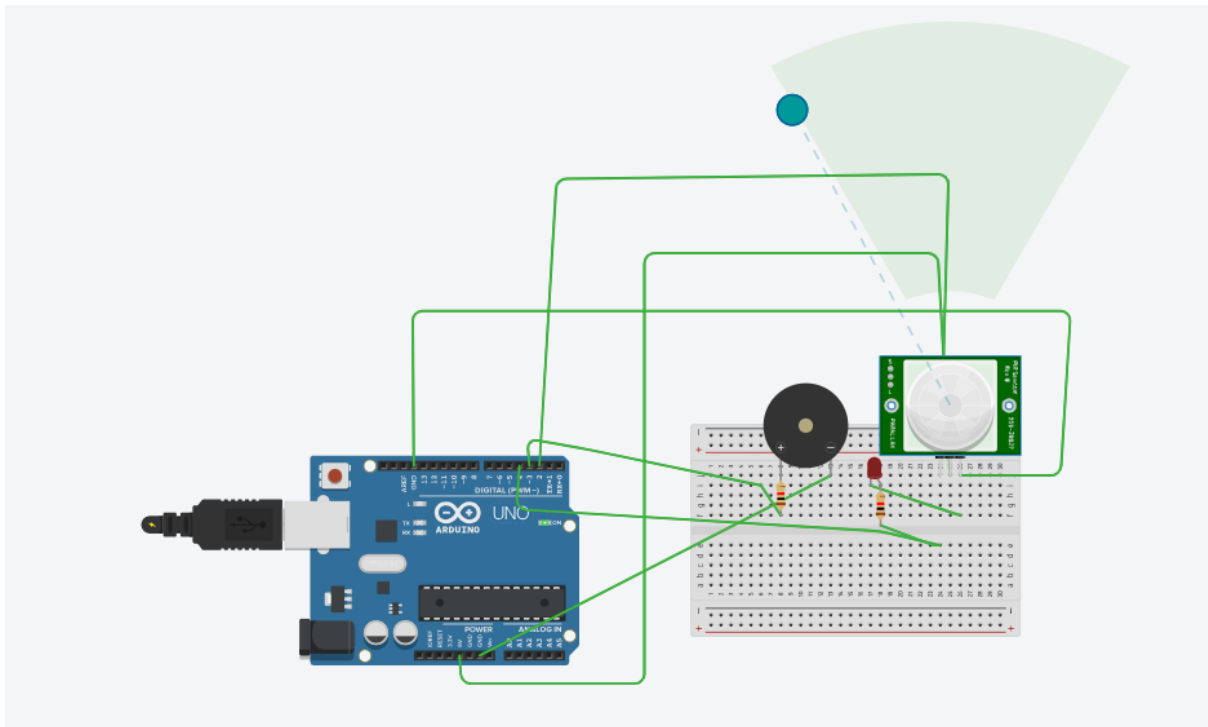
Buzzer with Resistor (100Ω – 1KΩ)

- Arduino Digital Pin 3 → One leg of 100Ω – 1KΩ resistor
- Other leg of resistor → Buzzer Positive (+)
- Buzzer Negative (-) → Arduino GND

PROCEDURE

- Gather all required components, including Arduino Uno, PIR sensor, LED, buzzer, resistors, jumper wires, and a breadboard.
- Connect the PIR sensor's VCC to Arduino 5V, GND to Arduino GND, and OUT to Arduino Digital Pin 2.
- Connect the LED's anode to a 220Ω – 1KΩ resistor and then to Arduino Digital Pin 4, while the cathode goes to GND.
- Connect the buzzer's positive terminal to a 100Ω – 1KΩ resistor and then to Arduino Digital Pin 3, while the negative terminal goes to GND.
- Verify all connections to ensure correct wiring and prevent short circuits before powering the Arduino.
- Test the system by moving in front of the PIR sensor to see if the LED lights up and the buzzer sounds.
- Adjust the PIR sensor's sensitivity and delay settings if needed to improve motion detection accuracy.
- If everything works fine, consider using an external power source and properly securing the components for real-world deployment.

CIRCUIT DIAGRAM:



CODE:

```
int motionSensorPin = 2;

int buzzerPin = 3;

int ledPin = 4;

int motiondetectedfreq = 2000;

int nomotionfreq = 500;

void setup() {

    pinMode(motionSensorPin, INPUT);
```

```
pinMode(buzzerPin, OUTPUT);
```

```
pinMode(ledPin, OUTPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
int motionState = digitalRead(motionSensorPin);
```

```
if (motionState == HIGH) {
```

```
tone(buzzerPin, motiondetectedfreq);
```

```
digitalWrite(buzzerPin, HIGH);
```

```
digitalWrite(ledPin, HIGH);
```

```
Serial.println("Motion Detected!");
```

```
}
```

```
else {
```

```
noTone(buzzerPin);
```

```
digitalWrite(buzzerPin, LOW);
```

```
digitalWrite(ledPin, LOW);
```

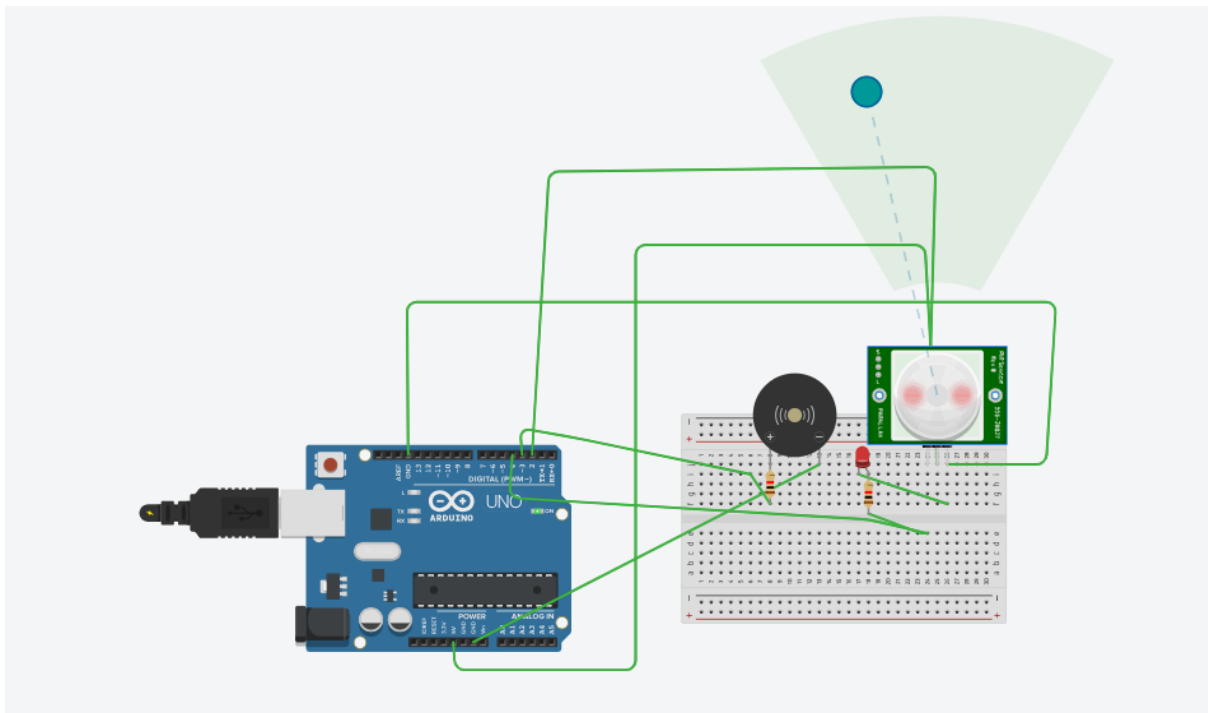
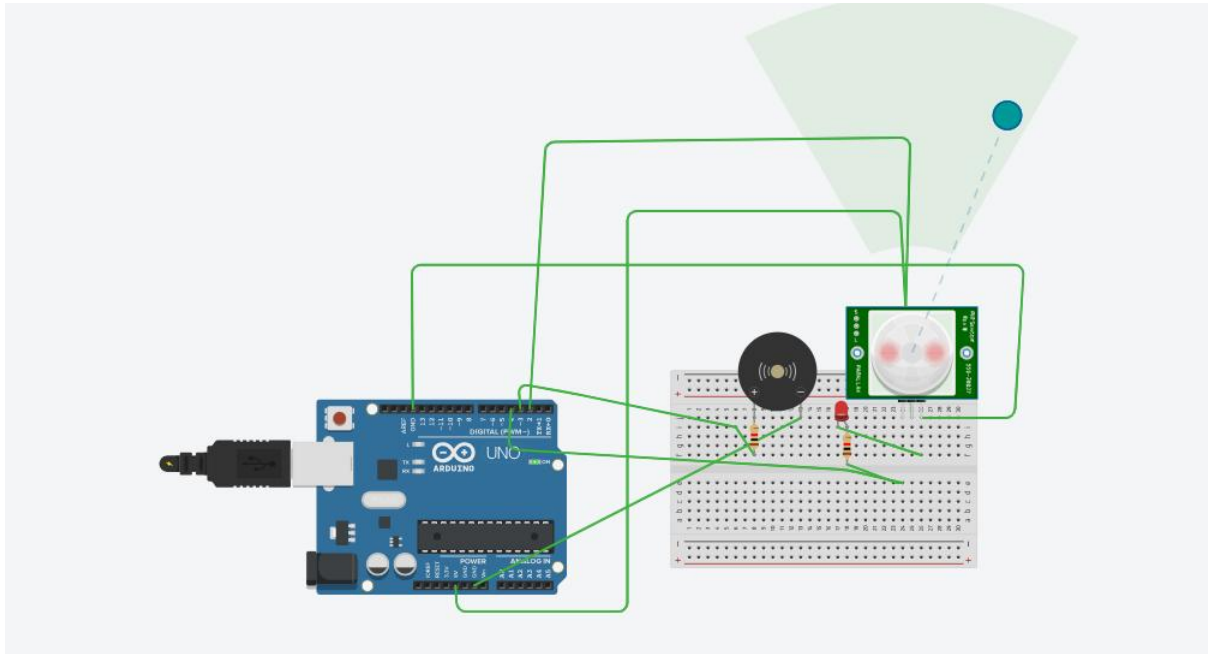
```
Serial.println("No Motion");
```

```
}
```

```
delay(100);
```

}

OUTPUT:



RESULT:

Hence the system built successfully monitors the change in the motion and the has a buzzer attached to it which emits a sound when the object is detected moving in its range. here the PIR motion sensor and the buzzer have been used to implement the system successfully and the results have been established.